Gabriel Ryan

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Research Summary

Research scientist working on AI-driven software development. My work spans model training and harness design, including completion, context retrieval, and agentic code review now shipping in GitHub Copilot to millions of developers. Before that, my PhD research combined program analysis and machine learning, developing neural methods for program verification, test generation, and vulnerability detection. I'm interested in post-training and agent harness design, and in automating the research and optimization loops behind them.

Education

Ph.D., Computer Science, Columbia University *(advised by Prof. Suman Jana)* Dec 2023
M.S., Computer Science, Columbia University Feb 2018
B.S. Engineering & B.A. Computer Science, Swarthmore College Jun 2013

Experience

Senior Research Scientist, Microsoft CodeAI Mar 2024 – Present

- Built automated optimization loops that orchestrate specialist subagents over multi-stage experiment campaigns, applied to the Copilot code review harness (+11pp F1 review comment improvement), LLM judges for online quality metrics (-43% error rate) and to embedding model training recipes (+10pp NDCG improvement).
- Post-trained agentic model for context retrieval subagent using RLVR and information-gain objectives with the Slime framework and SGLang for inference, achieving a 15% cost reduction in A/B testing on GitHub Copilot agent.
- Developed an efficient code embedding model used to index GitHub for Copilot's context retrieval served with vLLM, reducing indexing cost by $2\times$ and index size by $3\times$ while improving retrieval quality.
- Trained the code completion models powering GitHub Copilot in editors used by millions of developers, applying RLVR and reward models based on LLM judges and rubrics, fine-tuned on production usage data to drive a 5% acceptance-rate increase while reducing latency by 10%.
- Curate code datasets for mid-training and post-training, including synthetic data generation and SFT dataset construction.
- Configured and ran production A/B experiments through Microsoft's ExP platform, serving competing model variants to feed insights back into training and harness design.

Postdoctoral Research Scientist, Columbia ARiSE Lab Jan 2024 – Mar 2024

- Contributed to LLM-based linux kernel bug localization and repair research, including benchmark design and evaluation of LLMs on kernel bug repair tasks.

Doctoral Researcher, Columbia Security Lab Sep 2018 – Dec 2023

- Built a novel Linux kernel race prediction analysis using spectral learning, published in Oakland S&P and OOPSLA.
- Designed *Proximal Gradient Analysis* (PGA), a dataflow analysis method based on nonsmooth optimization. Implemented in LLVM. Published in USENIX Security.
- Designed the *Continuous Logic Network* (CLN), a neural architecture for logical learning and reasoning; applied it to learning program invariants for verification. Released a PyTorch library for constructing CLNs. Published in ICLR and PLDI.

Research Intern, AWS AI Labs Jun 2023 – Aug 2023

- Built a novel LLM regression-testing approach using static analysis to prompt the model to reason symbolically about program execution paths. Achieved $2\times$ coverage and $3\times$ correct test generations over baseline approaches on CodeGen2, GPT-3.5, and GPT-4. Published *Code-Aware Prompting* at FSE 2024.

Research Intern, Microsoft Research Jun 2021 – Aug 2021

- Built *TOGA: A Neural Method for Test Oracle Generation* using Transformers (CodeBERT) and a specialized grammar to generate unit tests, demonstrating a 170% improvement over prior work in bug detection.
- Published at ICSE 2022; awarded ACM SIGSOFT Distinguished Paper Award.

Software Engineer, Allure Security Technology Sep 2015 – Aug 2018

- Built systems for managing large-scale deployments of user-behavior sensors and detecting and tracking data loss across enterprise networks.

Robotics Software Consultant, 3DDataLtd May 2015 – Aug 2015

- Built a LiDAR- and inertial-sensor-based 3D mapping framework for robotic navigation.

Radar Software Engineer, Raytheon Sep 2013 – Apr 2015

- Built radar calibration, satellite tracking, and antenna diagnostic software; earned a team achievement award for delivering new diagnostic capabilities ahead of schedule.

Selected Publications

[ICML 2026] *SemRep: Generative Code Representation Learning with Code Transformations.* W. Li, J. Song, B. A. Stoica, A. Dhoot, **G. Ryan**, S. Fu, K. Pei.

[NeurIPS DL4C Workshop 2025] *DevBench: A Realistic, Developer-Informed Benchmark for Code Generation Models.* P. A. Golnari, A. Kumarappan, W. Wen, X. Liu, **G. Ryan**, Y. Sun, S. Fu, et al.

[OOPSLA 2024] *Accurate Data Race Prediction in the Linux Kernel through Sparse Fourier Learning.* **G. Ryan**, B. Cetin, Y. Lim, S. Jana. `https://dl.acm.org/doi/pdf/10.1145/3649840`

[FSE 2024] *Code-Aware Prompting: A study of Coverage Guided Test Generation in Regression Setting using LLM.* **G. Ryan**, S. Jain, M. Shang, S. Wang, X. Ma, M. Ramanathan, B. Ray. `https://arxiv.org/pdf/2402.00097`

[Oakland S&P 2023] *Precise Detection of Kernel Data Races with Probabilistic Lockset Analysis.* **G. Ryan**, A. Shah, D. She, S. Jana. `https://cs.columbia.edu/~gabe/files/oakland2023_pla.pdf`

[ICSE 2022] *TOGA: A Neural Method for Test Oracle Generation.* E. Dinella*, **G. Ryan***, T. Mytkowicz, S. Lahiri. `https://arxiv.org/pdf/2109.09262.pdf` **(Distinguished Paper Award)**

[OSDI 2021] *DistAI: Data-Driven Automated Invariant Learning for Distributed Protocols.* J. Yao, R. Tao, R. Gu, J. Nieh, S. Jana, **G. Ryan**. `https://www.usenix.org/system/files/osdi21-yao.pdf` **(Best Paper Award)**

[USENIX Security 2021] *Fine Grained Dataflow Tracking with Proximal Gradients.* **G. Ryan**, A. Shah, D. She, K. Bhat, and S. Jana. `https://arxiv.org/abs/1909.03461`

[PLDI 2020] *Learning Nonlinear Loop Invariants with Gated Continuous Logic Networks.* J. Yao*, **G. Ryan***, J. Wong*, S. Jana, and R. Gu. `https://arxiv.org/abs/2003.07959`

[ICLR 2020] *CLN2INV: Learning Loop Invariants with Continuous Logic Networks.* **G. Ryan***, J. Wong*, J. Yao*, R. Gu, and S. Jana. `https://arxiv.org/abs/1909.11542`

* denotes equal contribution.

Awards

ACM SIGSOFT Distinguished Paper Award, ICSE 2022 (TOGA).
Jay Lepreau Best Paper Award, OSDI 2021 (DistAI).
National Defense Science and Engineering Graduate Fellowship (NDSEG). Awarded NDSEG Fellowship for proposal ”Proximal Gradient Analysis for Vulnerability Detection and Defense.” 2019.
NSF Graduate Research Fellowship Honorable Mention. Accorded honorable mention for proposal ”Modeling and Simulating Adversarial User Behavior with Sequential VAEs.” 2018.

Academic Service

Workshop on Large Language Models for Code, ICSE 2026. Program Committee.
Next-Gen Code Development with Collaborative AI Agents Workshop, AAAI 2026. Program Committee.
AAAI 2024, 2025, 2026. Program Committee.

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL, Ada.

Frameworks & Infrastructure: PyTorch, Transformers, Hugging Face, Slime (RL framework), Ray, SGLang (RL inference), vLLM (evaluation serving), DeepSpeed, Azure ML.

Methods: LLM post-training (SFT, DPO, RLVR, RLAIF), reward design with LLM judges and rubrics, synthetic data generation, SFT dataset curation, retrieval and embeddings.